

PROBLEM SET 2—SOLUTIONS

ECO403/503—INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

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Instructions: Due on 22/10/2020; scan or type your work and submit through the link in Moodle. You are encouraged to work in groups but each student is required to submit her or his own work. If you worked with a group, please indicate the names of each member in your submission.

1. A monopolist can produce a good in different qualities. The cost of producing a unit of quality s is s^3 . Consumers buy at most one unit and have valuation function:

$$v(s, \theta) = \begin{cases} \theta s & \text{if they consume one unit of quality } s, \\ 0 & \text{if they do not consume.} \end{cases}$$

The monopolist decides on the quality (or qualities) it is going to produce and price. Consumers observe qualities and prices and decide which quality to buy if at all. Denote the price (or prices) that the monopolist charges the consumer by P .

- (a) Interpret the parameter θ of the consumers.
- (b) Characterize the first-best solution, assuming that θ belongs to a subset $\Theta \subset \mathbb{R}$.
- (c) How do the first-best quality depends on θ ?
- (d) Now assume that $\Theta = \{\theta_L, \theta_H\}$ with $\theta_L < \theta_H$. Furthermore, suppose that the seller cannot observe θ but believes that $\theta = \theta_H$ with probability β and that $\theta = \theta_L$ with probability $1 - \beta$. Write the optimization program that the monopolist will need to solve.
- (e) Characterize the second-best solution and the consumer's optimal informational rents.

Solution

- (a) Assume that $\theta \in \mathbb{R}_+$. Notice that $\frac{\partial v(s|\theta)}{\partial s} = \theta$ so that the marginal utility with respect to quality depends on the parameter θ . Thus, the larger the consumer's θ -parameter, the larger it is the marginal utility with respect to quality.

- (b) For the First-Best, recall that for each type, the principal solves

$$\begin{aligned} \max_{s_\theta, P_\theta} \{ & P_\theta - s_\theta^3 \} \\ \text{s.t. } & \theta s_\theta - P_\theta \geq 0 \end{aligned} \tag{1}$$

where (s_θ, P_θ) is a type-dependent menu of qualities and transfers. Note that the participation constraint will bind, i.e. $P_\theta = \theta s_\theta$ and thus, the principal can solve the program:

$$\max_{q(\theta)} \{ \theta s_\theta - s_\theta^3 \} \tag{2}$$

The first order condition for this program is

$$\theta = 3(s_\theta^f)^2 \Rightarrow s_\theta^f = \sqrt{\frac{\theta}{3}},$$

which characterizes the optimal quantity. The previous expression together with $P_\theta^f = \theta s_\theta^f$ characterize the optimal first-best contract.

- (c) For the First-Best quality

$$\frac{ds_\theta^f}{d\theta} = \frac{1}{2\sqrt{3\theta}} > 0$$

for every $\theta \in \mathbb{R}_+$. Thus quality is increasing with θ .

(d) The monopolist needs to solve the following program:

$$\begin{aligned}
& \max_{\{(s_L, s_H), (P_L, P_H)\}} \{\beta(P_H - s_H^3) + (1 - \beta)(P_L - s_L^3)\} \\
& \text{s.t.} \quad \theta_H s_H - P_H \geq \theta_H s_L - P_L \\
& \quad \theta_L s_L - P_L \geq \theta_L s_H - P_H \\
& \quad \theta_H s_H - P_H \geq 0 \\
& \quad \theta_L s_L - P_L \geq 0.
\end{aligned} \tag{3}$$

(e) In problem (3), the participation constraint for the low type and the incentive compatibility constraint for the high type imply the participation constraint for the high type. Furthermore, the ability of the high type to mimic the low type but not the other way around implies that the incentive compatibility constraint of the low type will be satisfied and thus can be omitted. Indeed, consider the first-best contract, which involves zero rents for both types, that contract is not incentive compatible because the θ_H buyer will prefer to choose the contract for the low type than her own first-best allocation: while this allocation restricts consumption, it nevertheless allows the high type to experience a positive informational rent. On the other hand, the low type θ_L will not find attractive to raise his consumption to the first best, since doing so would involve a transfer that, while leaving the high type with zero utility, would lead to a low type to experience a negative payoff since he has lower valuation for consumption.

Furthermore, the participation constraints for the low type and the incentive constraint for the high type will hold with equality:

$$\theta_H s_H - P_H = \theta_H s_L - P_L, \quad P_L = \theta_L s_L.$$

Solving for transfers and substituting them into problem (3) yields

$$\max_{(q_H, q_L)} \beta(\theta_H s_H - (\theta_H - \theta_L)s_L - s_H^3) + (1 - \beta)(\theta_L s_L - s_L^3). \tag{4}$$

The first order conditions for program (4) yield:

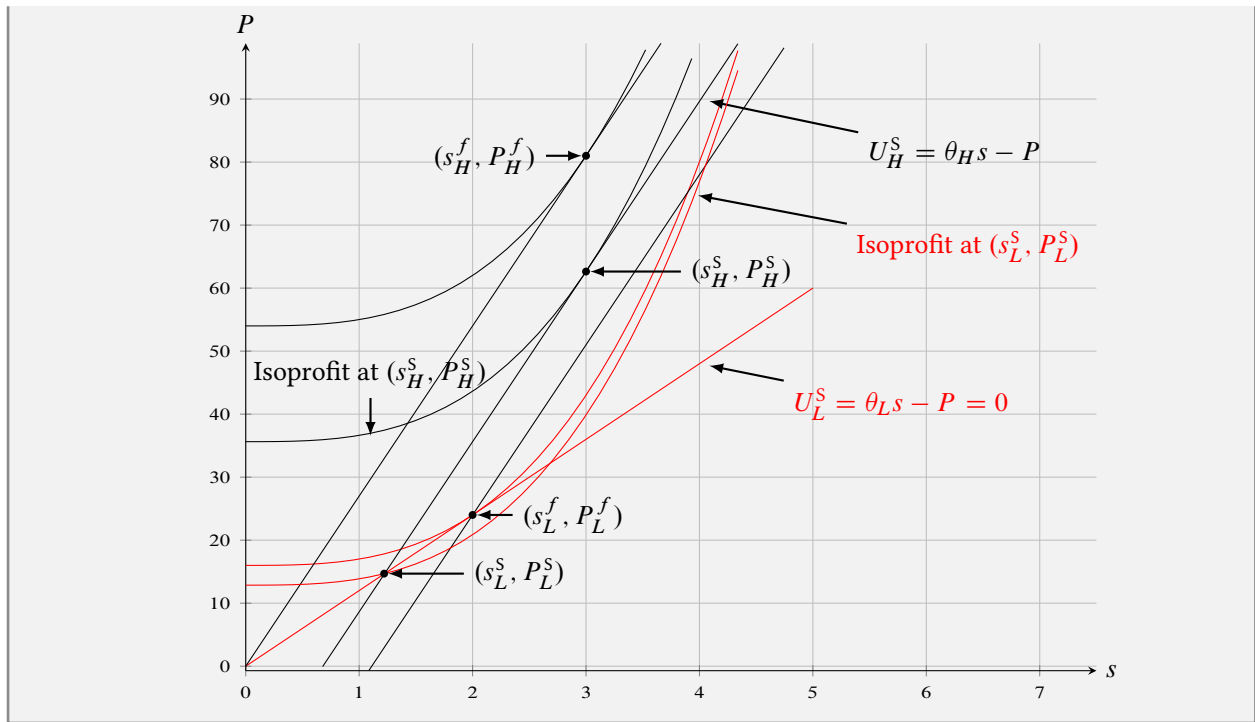
$$\begin{aligned}
& \theta_H - 3(s_H^S)^2 = 0 \Rightarrow s_H^S = \sqrt{\frac{\theta_H}{3}} \\
& -\beta(\theta_H - \theta_L) + (1 - \beta)(\theta_L - 3(s_L^S)^2) \leq 0 \Rightarrow s_L^S = \max \left\{ \sqrt{\frac{\theta_L}{3} - \frac{\beta}{3(1 - \beta)}(\theta_H - \theta_L)}, 0 \right\},
\end{aligned}$$

where the last equation accounts for the fact that s_L^S can have a corner solution, i.e., a binding non-negativity constraint. Prices are given by

$$\begin{aligned}
P_L^S &= \theta_L \max\{s_L^S, 0\}, \\
P_H^S &= \theta_H s_H^S - (\theta_H - \theta_L)s_L^S.
\end{aligned}$$

The informational rent for the low type agent is zero and for the high type agent is $(\theta_H - \theta_L)s_L^S \geq 0$ with strict inequality as long as s_L^S is strictly positive.

Here is a graph of the solution when $\theta_H = 27$, $\theta_L = 12$ and $\beta = 1/3$. With these parameter values, we have a strictly positive solution for s_L^S .



2. (J. Tirole) Assume that there are two types of consumers for a firm's product, θ_H and θ_L . The proportion of type θ_L consumers is λ . A type θ 's utility when consuming amount x of the good and paying a total T for it is $u(x, T) = \theta v(x) - T$, where

$$v(x) = \frac{1 - (1 - x)^2}{2}.$$

The firm is the sole producer of this good, and its cost of production per unit is $c > 0$.

- Consider a non-discriminating monopolist. Derive his optimal pricing policy, i.e. the cost per unit that the monopolist will charge. Show that he serves both classes of consumers if either θ_L or λ is "large enough." *Hint: Derive the demand curve for each type as a function of the price charged by the monopolist and then compute the "aggregate demand". Use this demand to solve for the optimal price of the monopolist.*
- Consider a monopolist who can distinguish the two types (by some characteristic) but can only charge a simple price p_i for each type θ_i . Characterize the optimal prices. Show that at the optimal prices, the monopolist is charging a marginal price $p = c$ per unit plus a personalized fixed fee equal to the consumer's *net surplus*.
- Suppose that the monopolist cannot distinguish the types. Suppose that the monopolist can charge a single *two-part tariff*. A two part tariff is an *affine pricing schedule*: $T(q) = A + pq$, where A is a fixed fee and p is a marginal price per unit. Derive the optimal two part tariff under the assumption that the monopolist serves both types.
- Compute the fully optimal nonlinear tariff (second best optimum for the monopolist's screening problem). How do the quantities purchased by the two types compare with the levels obtained in (a)–(c). How do the monopolist's profits from each case in (a)–(d) compare?

Solution

It will be useful to calculate some expressions that will be used repeatedly. For each type, the first order condition in the utility maximization problem of the consumer when he faces a *single-posted price* yields the individual demands:

$$\theta_i(1 - x_i) = p_i \Rightarrow x_i(p) = \max \left\{ 1 - \frac{p}{\theta_i}, 0 \right\} \quad \text{for } i = L, H. \quad (5)$$

Assume that $\theta_H > \theta_L > c > 0$ in order to simplify the analysis by eliminating the cases in which no

consumer would like to buy at the lowest price the monopolist can produce without incurring in losses. In order to simplify calculations, let θ denote the *harmonic mean* of θ_L and θ_H :

$$\frac{1}{\theta} \equiv \frac{\lambda}{\theta_L} + \frac{1-\lambda}{\theta_H}. \quad (6)$$

Then the aggregate demand at any price p can be written as

$$X(p) = \lambda x_L(p) + (1-\lambda)x_H(p) = 1 - \frac{p}{\theta}. \quad (7)$$

Net consumer surplus at a price p is given by

$$S_i(p) = \theta_i v(x_i(p)) - p x_i(p)$$

so that for the specific functional form that we assumed for the utility the previous expression yields

$$S_i(p) = \theta_i \left(\frac{1 - (1 - x_i(p))^2}{2} \right) - p x_i(p) = \frac{(\theta_i - p)^2}{2\theta_i}. \quad (8)$$

Note that $S_i \geq 0$ and that surplus is always higher for θ_H types.

- (a) The monopolist that can only charge a single price p would solve the following problem

$$\Pi_1 = \max_{p \geq 0} (p - c)X(p) \quad (9)$$

Since $X(p) \leq 0$ for any $p \geq \theta$ and the objective is strictly concave in p , the first order condition is necessary and sufficient to identify a solution to the monopolist's problem, which yields the monopoly price

$$p_1 = \frac{\theta - c}{2}$$

and the monopoly profits

$$\Pi_1 = \frac{(\theta - c)^2}{4\theta}.$$

Note that this solution assumes that the monopolist chooses to serve the two types of consumers. The monopolist could potentially choose a price high enough such type θ_L consumers choose not to buy. Such a strategy is optimal if the monopoly price relative to this category, which is $\frac{c+\theta_H}{2}$, exceeds θ_L and if the profits from this strategy are larger than the monopoly profits of serving the two types. Serving only the high type at monopoly price $(c + \theta_H)/2$ yields

$$\Pi'_1 = (1-\lambda) \frac{(\theta_H - c)^2}{4\theta_H}.$$

It can be shown that^a $\frac{\partial(\Pi_1 - \Pi'_1)}{\partial \lambda} > 0$. Furthermore, it is easy to see that $\lim_{\lambda \rightarrow 0} (\Pi_1 - \Pi'_1) < 0$ and that $\lim_{\lambda \rightarrow 1} (\Pi_1 - \Pi'_1) > 0$ and thus, by monotonicity and the intermediate value theorem, $(\Pi_1 - \Pi'_1)$ is equal to zero at just one point λ^* . Thus, for $\lambda \geq \lambda^*$, i.e. if λ is "high enough", serving the two types of consumers is more profitable than only serving the high types. Alternatively, if $\theta_L > (c + \theta_H)/2$, monopoly pricing for the type θ_H consumers yields consumption for the type θ_L consumers.

- (b) If the monopolist can distinguish between the two types then we are calculating the first-best outcome. Thus, the pair of problems of the perfectly discriminating monopolist are given by

$$\max_{x_i, T_i} T_i - c x_i \quad \text{subject to } \theta_i v(x_i) - T_i \geq 0 \text{ for } i = L, H.$$

As usual, the participation constraint must be binding and we can simplify the pair of problems to the following ones:

$$\max_{x_i, T_i} \theta_i v(x_i) - c x_i \quad \text{subject to } \theta_i v(x_i) - T_i \geq 0 \text{ for } i = L, H.$$

Notice that the monopolist is maximizing social surplus for each type. Due to the specific form for v that we assumed, the objective is strictly concave and continuous and therefore, the first order conditions are necessary and sufficient. From them we get

$$\theta_i(1 - x_i^f) = c \Rightarrow x_i^f = 1 - \frac{c}{\theta_i}, i = L, H.$$

At this solution, we have that $T_i^f = \theta_i v(x_i^f) = \frac{(\theta_i - c)(\theta_i + c)}{2\theta_i}$. Notice that this transfer is equivalent to the monopolist charging a marginal price of $p = c$ per unit and also charging a fixed fee equal to the net surplus of each consumer $S_i(c) = \frac{(\theta_i - c)^2}{2\theta_i}$. Notice that

$$S_i(c) + cx_i^f = \frac{(\theta_i - c)^2}{2\theta_i} + cx_i(p) = \frac{(\theta_i - c)^2}{2\theta_i} + \frac{\theta_i c - c^2}{\theta_i} = \frac{\theta_i^2 - c^2}{2\theta_i} = \frac{(\theta_i - c)(\theta_i + c)}{2\theta_i} = T_i^f.$$

Profits from the first best are

$$\Pi_2 = \lambda(T_L^f - cx_L^f) + (1 - \lambda)(T_H^f - cx_H^f) = \frac{\lambda(\theta_L - c)^2}{2\theta_L} + \frac{(1 - \lambda)(\theta_H - c)^2}{2\theta_H}.$$

- (c) Now we look for the optimal single two-part tariff. Again, for simplicity, we will assume that the monopolist serves the two types of consumers, i.e. that λ and θ_L are sufficiently large, as we saw before.

Suppose that the monopolist charges a marginal price of p . The highest fixed fee that is consistent with type θ_L consumers buying the good is $A = S_L(p)$. Type θ_H consumers then buy because as mentioned before $S_H(p) > S_L(p) = A$. Thus, the monopolist's problem is

$$\max_{p \in [0, \infty)} S_L(p) + (p - c)X(p).$$

The monopolist will do at least as well under the optimal single two-part tariff as under a fully linear tariff (single monopoly price). To the variable profit $(p - c)X(p)$ the monopolist is adding the fixed fee $S_L(p)$ that is paid by all types of consumers. Alternatively, notice that a linear tariff is a special case of a two-part tariff.

The first order condition yields

$$\frac{p - \theta_L}{\theta_L} - \frac{p}{\theta} + \frac{c}{\theta} + 1 - \frac{p}{\theta} = 0.$$

Solving the previous equation yields $p_3 = \frac{c}{2 - \frac{\theta}{\theta_L}}$. The monopolist profit, after a lot of tedious algebra is

$$\Pi_3 = \frac{1}{4} \left(c^2 \left(\frac{1}{\theta} + \frac{1}{2\theta_L - \theta} \right) - 4c + 2\theta_L \right).$$

- (d) Finally, for the optimal non-linear tariff, we follow the standard procedure. From the general problem analyzed in class, we know that the monopolist will solve

$$\max_{(x_i, T_i)_{i=L, H}} \lambda(T_L - cx_L) + (1 - \lambda)(T_H - cx_H) \quad (10)$$

$$\text{subject to } \theta_H v(x_H) - T_H \geq \theta_H v(x_L) - T_L \quad (11)$$

$$\theta_L v(x_L) - T_L \geq \theta_L v(x_H) - T_H \quad (12)$$

$$\theta_H v(x_H) - T_H \geq 0 \quad (13)$$

$$\theta_L v(x_L) - T_L \geq 0 \quad (14)$$

$$x_L \geq 0, x_H \geq 0 \quad (15)$$

This problem is equivalent to the reduced problem in which we drop the incentive constraint of the low type, the participation constraint of the high type and we impose a monotonicity constraint. Furthermore, the incentive constraint of the high type and the participation constraint of the low type

need to be binding. Using these facts and substituting the binding constraints into the objective and rewriting it we can solve the following problem:

$$\max_{x_L \geq 0, x_H \geq 0} \theta_L v(x_L) - cx_L + \frac{1-\lambda}{\lambda} (\theta_H v(x_H) - cx_H) - \frac{1-\lambda}{\lambda} (\theta_H v(x_L) - \theta_L v(x_L))$$

Notice that in the previous problem, as usual we have dropped the monotonicity constraint. We will check later that it holds.

From the first order conditions we get that

$$\theta_H (1 - x_H^s) = c \Rightarrow x_H^s = 1 - \frac{c}{\theta_H},$$

thus we have once again that there is no distortion at the top, and the second best quantity for the high type is the same as the first best.

For the low type:

$$\theta_L (1 - x_L^s) - c - \frac{1-\lambda}{\lambda} (\theta_H (1 - x_L^s) - \theta_L (1 - x_L^s)) = 0$$

which implies that

$$x_L^s = 1 - \frac{c}{\theta_L - \frac{1-\lambda}{\lambda} \Delta \theta}$$

where $\Delta \theta = \theta_H - \theta_L$. Notice that indeed $x_L^s \leq x_H^s$ and thus the monotonicity constraint is satisfied. The optimal transfers are

$$T_L^s = \theta_L v(x_L^s), \quad T_H^s = \theta_H (v(x_H^s) - v(x_L^s)) + \theta_L v(x_L^s).$$

Let Π_4 be the optimal non-linear tariff.

It is clear from the way we solved the problem, (more restricted to less restricted) that $\Pi_1 \leq \Pi_3 \leq \Pi_4 \leq \Pi_2$.

^a  **Warning!** The following part requires a ridiculous amount of algebra. But for the sake of completeness I will write here the proof that the derivative with respect to λ of the profit difference $\Pi_1 - \Pi'_1$ is strictly positive and thus increasing in λ . Calculating the derivative we get

$$\frac{\partial(\Pi_1 - \Pi'_1)}{\partial \lambda} = \frac{(\theta_H - \theta_L) \left(\frac{\theta_H \theta_L}{\lambda \theta_H - \lambda \theta_L + \theta_L} - c \right)^2}{4 \theta_H \theta_L} + \frac{(\theta_H - \theta_L) \left(\frac{\theta_H \theta_L}{\lambda \theta_H - \lambda \theta_L + \theta_L} - c \right)}{2 (\lambda \theta_H - \lambda \theta_L + \theta_L)}.$$

Expanding all terms, simplifying and canceling terms, then factoring and collecting similar terms yields:

$$\frac{1}{4} \left(c \left(\frac{c}{\theta_L} - 2 \right) - \frac{\theta_H^2 \theta_L}{(\lambda \theta_H - (\lambda - 1) \theta_L)^2} + \theta_H \left(\frac{\theta_L^2}{(\lambda \theta_H - \lambda \theta_L + \theta_L)^2} + 1 \right) \right).$$

We want to show that the previous expression is greater than zero to show that the difference in profits is strictly increasing with respect to λ . The proof will now proceed in two steps: we will show that when $\lambda \rightarrow 0$, the term that we obtain is strictly positive. Next we will show that the derivative is also an increasing function with respect to λ by calculating the second derivative and showing that it is positive. These two facts will allow us to conclude that the first derivative is strictly increasing and thus always strictly positive over its domain.

First note that when $\lambda \rightarrow 0$, the previous expression evaluates to

$$\frac{(c - \theta_H)(c + \theta_H - 2\theta_L)}{\theta_L}.$$

The previous term is strictly positive if and only if $c + \theta_H - 2\theta_L < 0$ which is exactly the condition that the monopoly price with respect to the type- θ_H market is less than the highest willingness to pay of type- θ_H . Now, calculating the second derivative with respect to λ yields:

$$\frac{\theta_H \theta_L (\theta_H - \theta_L)^2}{2 (\lambda \theta_H - \lambda \theta_L + \theta_L)^3},$$

which once again is easy to see is strictly positive. Hence, the first derivative is an increasing function and is strictly positive in all of its domain which concludes the proof.

- Let the agent be a drinker who plans to buy at most one bottle of wine within a month. His utility is $U = \theta q - t$ where q is the quality he buys and θ is a positive parameter that indexes his taste for quality. If he decides not to buy any wine, his utility is just 0. Suppose that there are n types of

drinkers and that $\theta_1 < \dots < \theta_n$. Their respective prior probabilities are $\pi_1, \dots, \pi_n$ with $\sum_{i=1}^n \pi_i = 1$. The principal is a local monopoly on the wine market. He can produce wine with any quality $q \geq 0$; the production of a bottle of wine with quality q costs him $c(q) = \frac{q^2}{2}$.

(a) Show that the only binding constraints are the downward adjacent incentive constraints, i.e.

$$\theta_i q_i - t_i \geq \theta_i q_{i-1} - t_{i-1} \quad i = 2, \dots, n$$

and the participation constraint for the lowest type, i.e.

$$\theta_1 q_1 - t_1 \geq 0$$

(b) Characterize as far as you can the solution to the monopolist's problem.

For the next part of the problem, assume that $n = 3$ and that $\theta_1 = 1, \theta_2 = 5, \theta_3 = 9$.

(c) Suppose that $\pi_1 = 0.85, \pi_2 = 0.08$ and that $\pi_3 = 0.07$. Compute the optimal qualities for each consumer, the optimal transfers and the monopolist's profits.

(d) Suppose now that $\pi_1 = \pi_2 = \pi_3 = 1/3$. Compute again the optimal qualities, transfers and the monopolist's profits.

(e) Compare the results from (c) and (d). What did you find? In which of the two cases is the monopolist performing better? Can you provide an economic interpretation of your results?

For the remainder of the problem, suppose now that the set of types is a continuum, specifically, assume that $\theta \in [\theta_L, \theta_H]$, where $0 < \theta_L < \theta_H$.

(f) Assume that θ is uniformly distributed on $[\theta_L, \theta_H]$. Characterize the second-best optimal quality-pricing schedule.

(g) Now suppose that θ is distributed on $[\theta_L, \theta_H]$ according to the distribution $f(\theta) = \frac{2(\theta_H - \theta)}{(\theta_H - \theta_L)^2}$. Characterize the second-best optimal quality-pricing schedule in this case. Is the hazard rate function $\frac{1-F(\theta)}{f(\theta)}$ decreasing? What does a decreasing hazard rate have to do with the monotonicity of the quality-pricing schedule?

Solution

(a) We start with an approach similar to the one in the lecture slides. The monopolist's problem is

$$\max_{\{(q_i, t_i)\}_{i=1, \dots, n}} \sum_{i=1}^n \pi_i \left(t_i - \frac{q_i^2}{2} \right) \quad (16)$$

subject to

$$\theta_i q_i - t_i \geq \theta_i q_j - t_j, \quad \forall i, j \quad (17)$$

$$\theta_i q_i - t_i \geq 0, \quad \forall i \quad (18)$$

$$q_i \geq 0, \quad \forall i \quad (19)$$

The previous problem is equivalent to the relaxed problem:

$$\max_{\{(q_i, t_i)\}_{i=1, \dots, n}} \sum_{i=1}^n \pi_i \left(t_i - \frac{q_i^2}{2} \right) \quad (20)$$

subject to

$$\theta_i q_i - t_i \geq \theta_i q_{i-1} - t_{i-1}, \quad \forall i = 2, \dots, n \quad (21)$$

$$\theta_1 q_1 - t_1 \geq 0, \quad (22)$$

$$q_i \geq q_{i-1}, \quad \forall i = 2, \dots, n \quad (23)$$

$$q_i \geq 0, \quad \forall i \quad (24)$$

First we will show that when monotonicity (23) holds, then all the incentive constraints (17) hold if

and only if the downward constraints (21) and the upward constraints

$$\theta_i q_i - t_i \geq \theta_i q_{i+1} - t_{i+1}, \forall i = 1, \dots, n-1, \quad (25)$$

hold. Clearly the full set of incentive constraints (17) imply both the downward (21) and upward (25) constraints. To prove the other direction, fix an index $i \in \{2, \dots, n\}$. Now we need to pick an arbitrary index j but as mentioned in class, we have two cases:

- i. $j < i$: This case was done in class and I am going to skip it. Follow the lecture notes to prove this case.
- ii. $j > i$: This is the case that I asked you to verify in the lecture notes. I am going to prove this case here.

Thus, take any $j > i$. Notice that

$$\theta_i q_i - q_j \theta_i = \sum_{k=1}^{j-1} \theta_i q_k - \theta_i q_{k+1} \geq \sum_{k=1}^{j-1} \theta_k q_k - \theta_k q_{k+1} \geq \sum_{k=1}^{j-1} t_k - t_{k+1} = t_i - t_j,$$

where the first equality is by a telescopic sum argument, the second inequality is by increasing differences, the third inequality is by upward incentive compatibility (25) and the fourth equality is again by a telescopic sum argument.

We continue with the proof. We proceed in 4 steps:

Step 1 The constraint set of P is a subset of RP. Notice that the full set of incentive compatibility restrictions (17) imply that

$$\theta_{i-1} q_i - \theta_{i-1} q_{i-1} \leq t_i - t_{i-1} \leq \theta_i q_i - \theta_i q_{i-1}$$

which implies $(q_i - q_{i-1})(\theta_{i-1} - \theta_i) \geq 0$. Since types are such that $\theta_i \geq \theta_{i-1}$ (this is the SCP in for this model), then we have that the previous inequality implies that $q_i \geq q_{i-1}$.

Step 2 At the solution to RP, (21) holds as an equality for all i and (22) also holds as an equality. Otherwise, we can increase all t_j , $j \geq i$ or t_1 by some $\epsilon > 0$, increasing the profit of the monopolist.

Step 3 Solutions to RP satisfy the constraints of P. Since (21) hold with equality and $q_i \geq q_{i-1}$, we have that

$$t_i - t_{i-1} = \theta_i q_i - \theta_i q_{i-1} \geq \theta_{i-1} q_i - \theta_{i-1} q_{i-1},$$

that is, all upward IC constraints (25) hold, and by the first result we proved here, this implies that the full set of IC constraints (17) hold.

Step 4 The participation constraints of the lowest type (22) implies all participation constraints (18). Note that the full set of IC (17), PC for the lowest type (22), and single crossing ($\theta_i \geq \theta_{i-1}$) imply that

$$\theta_i q_i - t_i \geq \theta_i q_1 - t_1 \geq \theta_1 q_1 - t_1 = 0.$$

Thus we have shown that at an optimal solution, the only relevant constraints are (21) and (22) and that these need to hold with equality, thus completing the proof.

- (b) We can follow the general development in the slides to characterize the solution. Foregoing the monotonicity constraint in the relaxed problem, we get:

$$q_n = \theta_n, \quad (26)$$

$$q_i = \max \left\{ \theta_i - \frac{1 - \Pi_i}{\pi_i} (\theta_{i+1} - \theta_i) \right\}, \quad (27)$$

where $\Pi_i = \sum_{k=1}^i \pi_k$. We can see from (26) that there is no distortion for the top type, and from (27) there will be a distortion. Furthermore, the expression in (27) takes into consideration the possibility of *corner solutions*, i.e. when the first order condition identifies a negative solution for i , then we put $q_i = 0$.

Note that in order for the solution to be monotonic, we would have to impose more restrictions on the parameters. Let $\Delta_i = \theta_{i+1} - \theta_i$ for every $i = 1, \dots, n-1$, and let $H_i = \frac{1 - \Pi_i}{\pi_i}$ for $i = 1, \dots, n$

- A *necessary and sufficient* condition for monotonicity is: $\frac{\Delta_i}{\Delta_{i+1}} \geq \frac{H_{i+1}}{1 + H_i}$ for $i = 1, \dots, n-1$.

- *Sufficient conditions* for monotonicity are that H_i should be decreasing with i and that also Δ_i should be decreasing with i . These two conditions are the ones that I mentioned in the last slide of the lecture: the inverse hazard rate should be decreasing in i (monotone hazard rate property) and that the difference in marginal values between types adjacent types should also be decreasing in i .

To characterize the transfers, we use the binding constraints (22) and (21):

$$t_1 = \theta_1 q_1, \quad (28)$$

$$t_i = \theta_i(q_i - q_{i-1}) + t_{i-1}, \quad i = 2, \dots, n. \quad (29)$$

(c) Quantities, transfers and profits are:

$$(q_1, q_2, q_3) = \left(\frac{5}{17}, \frac{3}{2}, 9 \right), \quad (30)$$

$$(t_1, t_2, t_3) = \left(\frac{5}{17}, \frac{215}{34}, \frac{1255}{17} \right), \quad (31)$$

$$\text{Profits} = \frac{1007}{340}. \quad (32)$$

(d) Quantities, transfers and profits are:

$$(q_1, q_2, q_3) = (0, 1, 9), \quad (33)$$

$$(t_1, t_2, t_3) = (0, 5, 77), \quad (34)$$

$$\text{Profits} = \frac{41}{3}. \quad (35)$$

(e) The monopolist performs better under (d) than in (c). Notice that the monopolist is shutting down consumption for the lowest type under scheme (d). Thus, at the cost of not contracting with the lowest type, the monopolist is able to extract more rent from higher types, and this trade-off is optimal.

Define the *cumulative distribution functions* for each case as

$$G(\theta) = \begin{cases} 0 & \theta < \theta_1 \\ 0.85 & \theta_1 \leq \theta < \theta_2 \\ 0.93 & \theta_2 \leq \theta < \theta_3 \\ 1 & \theta_3 \leq \theta \end{cases} \quad F(\theta) = \begin{cases} 0 & \theta < \theta_1 \\ 1/3 & \theta_1 \leq \theta < \theta_2 \\ 2/3 & \theta_2 \leq \theta < \theta_3 \\ 1 & \theta_3 \leq \theta \end{cases}$$

Note that $F(\theta) \leq G(\theta)$ for every θ , so that $G \succ_{\text{fbsd}} F$. Because of this, the monopolist is confident that by shutting down contracting with the lowest type under F is profitable.

The remaining part is a monopoly screening problem with a continuum of types just like in the lecture notes. I will solve the problem first for an arbitrary distribution $f(\theta)$. The monopolist problem is

$$\begin{aligned} \max_{(q,t)} \quad & \int_{\theta_L}^{\theta_H} \left(t(\theta) - \frac{q(\theta)^2}{2} \right) f(\theta) d\theta \\ \text{subject to} \quad & \theta q(\theta) - t(\theta) \geq 0 \quad \forall \theta \in [\theta_L, \theta_H] \\ & \theta q(\theta) - t(\theta) \geq \theta q(\theta') - t(\theta') \quad \forall \theta, \theta' \in [\theta_L, \theta_H] \end{aligned}$$

For any contract (q, t) define $U(\theta) = \theta q(\theta) - t(\theta)$, i.e., the utility from the consumer of choosing a contract designed for his true type. A contract (q, t) is incentive compatible if and only if

- q is monotone increasing
- $U(\theta) = U(\theta') + \int_{\theta'}^{\theta} q(\tau) d\tau$.

You can check my slides for the proof of why the two conditions above are equivalent to incentive compatibility. Note that the above conditions imply that the only participation constraint that matters is the one for the lowest type θ_L . Once again, for the proof you can refer to the slides.

The problem of the monopolist is equivalent to

$$\begin{aligned}
& \max_{(q,t)} \int_{\theta_L}^{\theta_H} \left(t(\theta) - \frac{q(\theta)^2}{2} \right) f(\theta) d\theta \\
& \text{subject to } U(\theta) = U(\theta_L) + \int_{\theta_L}^{\theta} q(\tau) d\tau \\
& q \text{ is increasing} \\
& U(\theta_L) \geq 0.
\end{aligned}$$

Note that we can write using the envelope condition the transfers as $t(\theta) = \theta q(\theta) - U(\theta_L) - \int_{\theta_L}^{\theta} q(\tau) d\tau$. We can use this fact to rewrite the objective function, which after a lengthy simplification (in the slides, it is the part that involves simplifying the double integral that appears) yields the following problem for the principal

$$\begin{aligned}
& \max_q \int_{\theta_L}^{\theta_H} \left(\theta q(\theta) - \frac{q(\theta)^2}{2} - \frac{1 - F(\theta)}{f(\theta)} q(\theta) \right) f(\theta) d\theta \\
& \text{subject to } q \text{ increasing}
\end{aligned}$$

As usual, we will solve the problem by first ignoring the monotonicity condition and then identifying conditions under which it will hold.

We can maximize the above integral by choosing q such that it maximizes *pointwise* the integrand, which yields the first order condition:

$$\theta - q(\theta) - \frac{1 - F(\theta)}{f(\theta)} = 0.$$

Solving for q yields

$$q(\theta) = \max \left\{ \theta - \frac{1 - F(\theta)}{f(\theta)}, 0 \right\},$$

where the max takes care of the possibility of negative solutions. Define θ^* as the value that solves the following equation:

$$\theta^* = \frac{1 - F(\theta^*)}{f(\theta^*)}.$$

Then we can express $q(\theta)$ as

$$q(\theta) = \begin{cases} \theta - \frac{1 - F(\theta)}{f(\theta)} & \theta \in [\theta^*, \theta_H] \\ 0 & \theta \in [\theta_L, \theta^*] \end{cases}$$

Thus, notice that the contract exhibits shutdown for some types.

To calculate the transfers we first compute $U(\theta)$:

$$U(\theta) = \int_{\theta_L}^{\theta} \max \left\{ \tau - \frac{1 - F(\tau)}{f(\tau)}, 0 \right\} d\tau = \begin{cases} 0 & \theta \in [\theta_L, \theta^*] \\ \int_{\theta^*}^{\theta} \left(\tau - \frac{1 - F(\tau)}{f(\tau)} \right) d\tau & \theta \in [\theta^*, \theta_H] \end{cases}$$

Thus, transfers are $t(\theta) = \theta q(\theta) - U(\theta)$ where q and U are given by the formulas above.

(f) For the uniform distribution we have that

$$\begin{aligned}
f(\theta) &= \frac{1}{\theta_H - \theta_L} \\
F(\theta) &= \frac{\theta - \theta_L}{\theta_H - \theta_L} \\
\frac{1 - F(\theta)}{f(\theta)} &= \theta_H - \theta.
\end{aligned}$$

Thus $\theta^* = \frac{\theta_H}{2}$ so that

$$q(\theta) = \begin{cases} 0 & \theta \in [0, \theta_H/2] \\ 2\theta - \theta_H & \theta \in [\theta_H/2, \theta_H] \end{cases}$$

$$t(\theta) = \begin{cases} 0 & \theta \in [0, \theta_H/2] \\ \theta^2 - \frac{\theta_H^2}{4} & \theta \in [\theta_H/2, \theta_H] \end{cases}$$

(g) For this case we have

$$f(\theta) = \frac{2(\theta_H - \theta)}{(\theta_H - \theta_L)^2}$$

$$F(\theta) = \frac{2\theta_H\theta - \theta^2 - 2\theta_H\theta_L + \theta_L^2}{(\theta_H - \theta_L)^2}$$

$$\frac{1 - F(\theta)}{f(\theta)} = \frac{\theta_H - \theta}{2}.$$

Thus $\theta^* = \frac{\theta_H}{3}$ so that

$$q(\theta) = \begin{cases} 0 & \theta \in [0, \theta_H/3] \\ \frac{3\theta - \theta_H}{2} & \theta \in [\theta_H/3, \theta_H] \end{cases}$$

$$t(\theta) = \begin{cases} 0 & \theta \in [0, \theta_H/3] \\ \frac{9\theta^2 - \theta_H^2}{12} & \theta \in [\theta_H/3, \theta_H] \end{cases}$$

Notice that the hazard rate for this case is decreasing. Recall that a decreasing hazard rate is part of the sufficient conditions required for monotonicity of q . For the case in this item, q is indeed increasing in the type.